DApp Hybrid TEE-Rollup DA

[1]

1. 200062

DApp Hybrid TEE-Rollup DA DA DA Python EVM gas Sepolia 406 DA 96.33% TEE DA Hybrid TEE-Rollup DApp

A TP311

Recoverable Challenge Protocol and Lightweight DA Cost Evaluation for Hybrid TEE-Rollup in High-Frequency DApps

YANG Fan[1]

1. School of Data Science and Engineering, East China Normal University, Shanghai 200062, China

Abstract: High-frequency decentralized applications (DApps) require low latency, low on-chain overhead, and publicly verifiable arbitration. This paper proposes a recoverable challenge mechanism, an evidence-carrying compact commitment mechanism, and a data-availability-aware cost model for Hybrid TEE-Rollup. The recoverable challenge mechanism restores timeout sessions to replayable and settleable states; the compact commitment binds the state root, output hash, proof hash, DA pointer, and DA root; and the cost model analyzes payload size, batching amortization, and DA path selection. Based on a Python prototype, local EVM gas measurements, and Sepolia deployment records, the compact commitment averages about 406 bytes, the modular DA sampling path reduces amortized cost by about 96.33% under the given configuration, and the recovery path improves challenge completion under timeout scenarios. The conclusions are based on a simulated TEE, a verifiable DA registry, and a configurable cost model.

Key words: Hybrid TEE-Rollup; interactive challenge; data availability; high-frequency DApp; cost amortization; fault recovery

1

AI

TEE Hybrid TEE-Rollup TEE [1]

DApp

TEE DA EVM gas Sepolia

1

2 DA DA

3 DA DA DA

2

TEEROLLUP TEE Rollup

opML optimistic fraud proof FPVM ML Hybrid TEE-Rollup Dynamic Fraud Proof

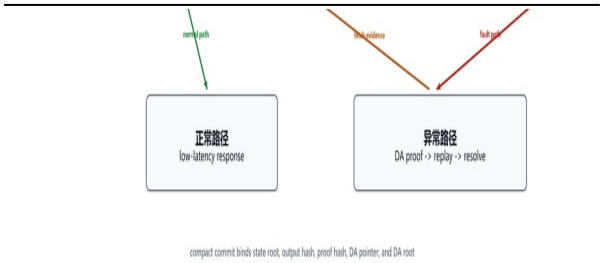
LazyLedger DA DA Rollup [2] ZK-Rollup Optimistic Rollup

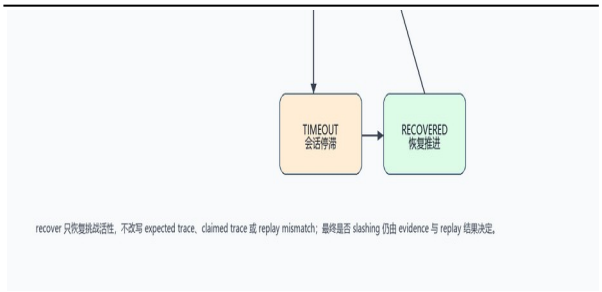
[3] Rollup [4][5]

[6]

Rollup [7][8][9][10]

Hybrid TEE-Rollup





DA

Merkle
TEE

3

DA
DA
AI

1 Hybrid TEE-Rollup
Fig.1 Prototype architecture of Hybrid TEE-Rollup

1 DApp

3.1

DApp
nonce
[11]
“ _ _ ”

3.2

DA
Merkle Merkle

Merkle DA Merkle

3.3

[4][5]
DA

Solidity MockTEEVerifier
MockDARegistry MerkleVerifier HybridTEERollup
EVM Python

4

4.1

4.1.1 协议模型与形式化定义

1
cs = (id, s, F, T)
id s F T

2
M = (S, E, δ)
S E δ: S × E → S

3
Recover(cs) = cs'
F(cs) = F(Recover(cs))

TIMED_OUT

4.1.2 超时、恢复、重放与仲裁结算

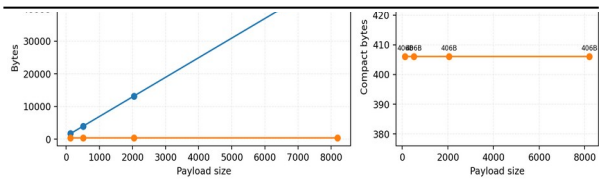
2
Fig.2 Workflow of the recoverable challenge protocol

n
log2(n)

“ ”

4.1.3 正确性边界与协议性质

DA Merkle DA
Merkle
Merkle DA DA



“

” “ ”

n

log2(n) 4 8 16 32 64 128

2 3 4 5 6 7

4.1.4 协议性质

“ ”

挑战事实不变性

“ ”

- State Root
- DA Merkle Root DA
- Payload Hash DA
- Execution Trace
-
- DA

cs Facts(cs)

Facts(cs_before) = Facts(cs_after)

安全性与活性分离

——

——

DA Merkle DA

仲裁连续性

—— DA

“ ”

“ ”

4.2

Merkle

4.3

DA DA

EIP-4844 DA [12]

C_total = C_commit + C_DA + p_challenge × C_challenge C_commit C_DA DA C_challenge p_challenge C_amortized = C_total / batch_size

5

Python TEE DA EVM gas Sepolia RQ1 RQ2 DA RQ3 RQ4 RQ5 Solidity EVM Sepolia

C_total = C_commit + C_DA + p_challenge × C_challenge 1

C_amortized = C_total / batch_size 2

C_commit C_DA DA C_challenge p_challenge

5.1

RQ1 Python TEE JSON

406

8192 64 256

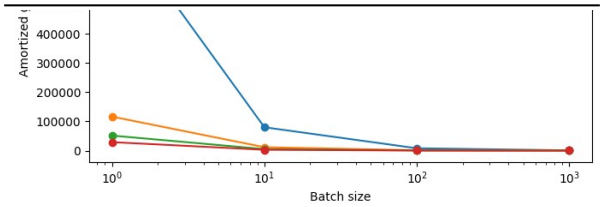
50068.84 50262.84 406.00

3

Fig.3 Byte-size comparison between full payloads and compact commits

5.2 DA

RQ2
DA EIP-4844 DA
gas
8192
gas 1 801101.44 1000
801.21 DA 29026.98 29.03 96.33%
DA



4 gas
Fig.4 Amortized gas trends under different DA configurations

5.3

RQ3 4.1.4 Python
4 128
2

100% 0%
100%
5 EVM challengeStep gas
93807 challengeRecover gas 156891
1.67

EVM

5.4

RQ4

0%

4
Table 4 Failure scenarios and evidence boundaries

□□	□□	□□□□	□□□□
	, ,	100%	0%
+	,	100%	
	vs	0%	100%

5.5 Solidity EVM Sepolia

RQ5 Solidity
Hardhat EVM gas 5
Sepolia Chain ID 11155111
10825904

5 EVM gasUsed
Table 5 Average gasUsed of local EVM key paths

	gasUsed	
registerDA	113526	MockDARegistry
submitRollup	299033.	
challengeOpen	283769	
challengeRespond	65886	
challengeStep	93807	
challengeRecover	156891	
challengeReplay	135632	
challengeResolve	113402	
challengeFinalize	33477	

67

Python
Hardhat

5.6

RQ1 RQ5
406 DA

6

Hybrid TEE-Rollup

DA DA
DA
Python EVM gas Sepolia
DA
TEE
DA
TEE

[1] WEN X, FENG Q, LYU H, et al. TEEROLLUP: efficient Rollup design using heterogeneous TEE[EB/OL]. arXiv:2409.14647, 2024 [2026-06-15].
[2] BUTERIN V. An incomplete guide to Rollups[EB/OL]. 2021 [2026-06-15].
[3] ETHEREUM FOUNDATION. Optimistic Rollups[EB/OL]. 2024 [2026-06-15].
[4] ARBITRUM FOUNDATION. Arbitrum Nitro technical documentation[EB/OL]. 2024 [2026-06-15].
[5] OP LABS. Optimism documentation: fault proofs and dispute games[EB/OL]. 2024 [2026-06-15].
[6] PICCO G, FORTUGNO A. Dynamic fraud proof[EB/OL]. arXiv:2502.10321, 2025 [2026-06-15].
[7] AL-BASSAM M. LazyLedger: a distributed data availability ledger with client-side smart contracts[EB/OL]. arXiv:1905.09274, 2019 [2026-06-15].
[8] TAS E N, TSE D, YANG L, et al. Light clients for lazy blockchains[EB/OL]. arXiv:2203.15968, 2022 [2026-06-15].
[9] EIGENLABS. EigenDA documentation[EB/OL]. 2024 [2026-06-15].
[10] CELESTIA. Celestia: modular blockchain network[EB/OL].

2024 [2026-06-15].
[11] INTEL CORPORATION. Intel Software Guard Extensions developer guide[EB/OL]. 2024 [2026-06-15].
[12] ETHEREUM FOUNDATION. EIP-4844: shard blob transactions[EB/OL]. 2024 [2026-06-15].

3663 200062
2991777340@qq.com
13319693336